Yashraj Singh Thakur

+91 7000501987 Mail LinkedIn Github Portfolio Leetcode

Education

Narmada Valley International School, Narmadapuram **2019 – 2020**
Class 10th *92.8%*

Narmada Valley International School, Narmadapuram **2021 – 2022**
Class 12th *97.4%*

Indian Institute of Technology, Patna **2023 – 2027**
B.Tech in Engineering Physics *8.36/10 CGPA*

Work Experience

AlgoUniversity (backed by Y-Combinator) **May 2025 – June 2025**
SWE Co-Op Intern *Remote*

- Built a code evaluation system supporting **3+ languages** and **100+ concurrent users**.
- Processed and analyzed logs from **5,000+ submissions** to identify execution delays.
- Reduced average response latency by **25%** through queue optimization and caching.
- Implemented authentication for **40+ users** with role-based access control.
- Standardized runtime environments using Docker to improve deployment stability.
- Collaborated with instructors to test new features and collect usage feedback.

Projects

Cohort – Online Competitive Programming Judge GitHub Deployment

- Handled **1,000+ submissions** using a Redis-based task queue under peak traffic.
- Monitored traffic from **500+ users** and improved throughput by **20%**.
- Reduced manual verification effort by **70%** using automated evaluation workflows.
- Deployed services on AWS EC2 with **99.9% uptime**.
- Implemented logging and alerts for failure detection and recovery.
- **Tools:** Node.js, MongoDB, Redis, AWS EC2, Vercel.

Algorithmic Trading Strategy GitHub

- Analyzed **10,000+ historical price points** to generate buy/sell signals.
- Backtested strategy across **3 market conditions** to compare performance trends.
- Stored financial data in PostgreSQL for repeatable experiments.
- Containerized services to enable reproducible deployments across machines.
- Measured system reliability using **82% test coverage**.

Smart Energy Forecast Dashboard *(FastAPI, ML, RL, PostgreSQL)* GitHub Link

- Trained forecasting models on **50,000+ consumption records** to predict demand.
- Evaluated model accuracy using historical validation sets and error metrics.
- Reduced estimated energy cost by **15–20%** through optimized scheduling.
- Tracked performance using dashboards with **10+ metrics**.
- Applied reinforcement learning to improve utilization of energy resources.
- **Tech:** FastAPI, XGBoost, Prophet, PostgreSQL, Chart.js.

Technical Skills

Programming: Python, Java, JavaScript, C

Data & Systems: SQL, PostgreSQL, MongoDB, Redis, Docker, AWS, GCP

Frameworks: Node.js, Express.js, FastAPI, React

Tools: GitHub, Postman, VS Code